MFSK

1. For MFSS, with $f_c = 250kHz$, $f_d = 25kHz$ and M=8, list all the frequency assignments for each of the data combinations. What is the data rate?

$f_i = f_c + (2i - 1 - M) f_d$

$= 250k + (2i - 9) \cdot 25\,kHz$

$1 \le i \le 8 = M$

$f_1 = 75\,kHz$

$f_2 = 125\,kHz$

175

225

275

325

375

$f_8 = 425\,kHz$

$R = \frac{1}{T} = 2L f_d = 150\,kbps$

$R = \frac{1}{T_S} = \frac{\log_2 M}{T_S} = 2 f_d \log_2 M$

$= 50 \times 3$

$= 150\,kbps$

2. A system transmits at 30 kbps, sending 3 bits per symbol. The time between hops for a FHSS system is 0.125ms. Is the system using slow-frequency-hop spread spectrum or fast-frequency-hop spread spectrum?

R L Tc

$T_c < T_s$

$T_c > T_s$

$R = 30\text{kbps}$ $L = 3$ $T_c = 0.125\text{ms}$

$R = \frac{L}{T_s} \Rightarrow T_s = \frac{L}{R} = \frac{3}{30k} = 0.1\text{ms} < T_c$

$\therefore$ slow

$R = \frac{L}{T_s}$

4. If the received signal level for a particular digital system is -151dBW and the receiver system effective noise temperature is 1500 K, what is E_b/N_0 for a link transmitting 2400bps?

$$\left(\frac{E_b}{N_0}\right)_{dB} = S_{dBW} - 10\log R - 10\log k - 10\log T \ dBW$$

$$= -151\,dBW - 10\log_{10} 2400 + 228.6\,dBJ/K - 10\log_{10} 1500$$

$$= 12\,dBW$$

5. In a simple free-space radio propagation model, the received signal power is proportional to $1/d^4$, where d is distance. Calculate the interfering power from the co-channel cells in a 7-cell cluster (P_{i7}) and compare it with the interfering power in a 3-cell cluster – i.e. evaluate P_{i7}/P_{i3} in dB. Assume the cell radius is the same in each case.

$R_u = r\sqrt{3N}$

6 surrounding cells

$$P_i = \frac{6P}{d^4} = \frac{6P}{R_u^4} = \frac{6P}{r^4 9N^2}$$

$$\frac{P_{i7}}{P_{i3}} = \frac{3^2}{7^2} = \frac{9}{49}$$

$= -7.4$ dB